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Parallel VLSI Architectures for High-Throughput Image Processing in Medical Imaging

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Abstract

The need for efficient and reliable image processing techniques has never been greater in medical imaging. The proposed work analyzes and designs parallel VLSI (Very Large-Scale Integration) architectures that meet the stringent demands of high-throughput medical image processing. The remainder of the paper is organized as follows, beginning with a discussion of the computational challenges that arise organically in medical imaging and why it is crucial to surmount these obstacles. The section on the proposed work describes the architecture, parallelization strategies, and processing pipeline, providing context for the hardware and Software underlying the system. The evaluation results below demonstrate significant enhancements in processing speed and image quality. Due to their real-time capabilities and high efficiency, parallel VLSI architectures have the potential to revolutionize medical image processing, resulting in more accurate diagnoses and more effective treatments. Research into optimizing these structures and discovering new applications is anticipated to result in further advances in medical imaging.

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Keywords: Parallel Processing; Processing Pipeline; Image Processing; Computational Demands

1. Introduction

Medical imaging has rapidly risen to the vanguard of modern medicine due to its capacity to detect and diagnose everything from minor diseases to potentially fatal conditions [1]. Digital radiography and medical imaging

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technologies such as computed tomography (CT), magnetic resonance imaging (MRI), and digital radiography generate tremendous quantities of digital image data [2].

Even though these innovations have unquestionably improved the accuracy and scope of medical diagnosis, they have also introduced significant computing challenges in processing and analyzing the vast quantities of visual data collected daily [3]. In the medical imaging industry, high-throughput image processing is essential due to the need for real-time diagnosis and treatment planning. Consequently, technology has substantially altered how medical professionals study diseases, identify symptoms, and monitor patients [4]. Medical imaging has enabled clinicians to gain unprecedented insight into the human body without invasive procedures. This capability has altered the face of medicine by enabling earlier detection, more precise diagnosis, and more individualized treatment for each patient.

CT scans and MRIs are widely used due to the extensive anatomical and functional information that can provide [5]. In contrast to computed tomography (CT) scans, which provide cross-sectional views of the body, magnetic resonance imaging (MRI) acquires high-resolution images without the use of ionizing radiation [6]. These imaging techniques are essential for detecting a vast array of medical conditions, including bone fractures, cancer, neurological disorders, and cardiovascular diseases. Digital radiography and ultrasonography are frequently used for rapid evaluations and monitoring in emergency and critical care contexts [7]. Despite the undeniable benefits of medical imaging, the vast volume and intricacy of image data present significant obstacles. Digital medical image archives have grown exponentially since the transition from analog to digital imaging, necessitating efficient storage, retrieval, and administration solutions. Nonetheless, the computer requirements for evaluating these photographs present a more immediate obstacle [8].

Medical image processing places significant demands on computer systems. Interventional radiology and other life-or-death applications require real-time processing above all else. Second, image enhancement, segmentation, and feature extraction methods have grown more complex and time-consuming. Third, the processing effort required to work with volumetric datasets generated by 3D and 4D imaging modalities, such as 3D CT scans and dynamic MRI, is several orders of magnitude more than that required to work with conventional 2D images [9]. Keeping up with the need for real-time processing is one of the greatest obstacles in medical image processing. During medical procedures such as surgery or interventional radiology, prompt access to processed image data is essential for accurate decision-making and a successful outcome. If photos are analyzed too slowly, it could result in inconvenient patient delays, increased radiation exposure, or even medical errors. Real-time medical image processing requires a substantial computational capacity and a processing pipeline with minimal latency [10].

Due to the increasing volume and complexity of medical image collections, conventional computer systems may not be able to meet these real-time demands, even if this is capable of performing medical image processing tasks. This research examines the design and evaluation of parallel Very Large-Scale Integration (VLSI) architectures to address these issues. These configurations are intended to maximize the processing efficacy of medical images using parallel processing. The remaining sections of this paper elucidate the proposed VLSI architecture, the parallelization techniques employed, the internal data flow, and the performance outcomes attained through meticulous testing. The proposed work aims to demonstrate that parallel VLSI architectures have the potential to significantly enhance medical image processing by facilitating real-time capabilities, bolstering computational efficiency, and enhancing diagnostic accuracy.

2. Literature Review

W. Zhou et al. [11] proposed that very large-scale integration (VLSI) hardware architecture is presented to implement HEVC (High-Efficiency Video Coding) deblocking filtering in real-time at high throughput. This design's unique technique for determining boundary conditions greatly simplifies implementation. It can simultaneously conduct luma/chroma processing and edge processing in both directions due to its four parallel filtering units. Compared to prior HEVC implementations, the experimental results disclose significant performance enhancements, albeit at a slightly higher hardware cost. Notably, the proposed architecture can achieve a staggering 123 frames per second (fps) at a clock rate of 278MHz, readily meeting the real-time requirements of deblocking filters for 8K and 4K video formats. This innovation is a significant contribution to video compression and processing because it can increase the efficacy with which ultra-high-definition content can be managed.

C. K. Chan et al. [12] described that constructing a high-sample-rate, wideband radar system, which is essential for advancing radar technology. Implemented in hardware as a 4-parallel 64K-point FFT, it is a novel variation on the conventional 2-epoch FFT method. If this generates twiddle factor coefficients in the middle, this can significantly reduce the quantity of ROM space required to store them. Due to the use of TSMC's 90 nm CMOS technology and a high-Vt standard cell library, the total number of gates is approximately 3974.1k. The greatest frequency of the system is 390 MHz, and the maximum throughput is 1.57 GS/s. This level of performance is accomplished while consuming 0.2811 W of power at 0.9 V. The proposed work considerably advances the state of the art in radar technology by combining real-time processing with ingeniously designed hardware.

Z. Shao et al. [13] proposed a revolutionary VLSI design for a high-throughput Huffman encoder that employs the Canonical Huffman technique to improve encoding performance and reduce the construction time for the Huffman code word table. The architecture employs parallel computing methodologies for frequency-statistical and code-size computational sorting to complete the tree-building and symbol-assigning processes with a single data scan. The time and effort savings are significant compared to the conventional two-stage data scanning technique. The architecture maintains the excellent compression ratio of Canonical Huffman while excelling in parallel processing capabilities. Compared to conventional algorithm-based Huffman encoder, testing reveals a 26.30 percent reduction in encoding time for 256 8-bit symbols.

P. Bhowmik et al. [14] state that bio-inspired computing is a method for translating sequential image processing into parallel computational processing units that can operate on the sensor. This research presents a hardware architecture for extracting data from an image. These processing nodes are organized in a vertically integrated hierarchical plane and feature a region-based Attention Module that isolates the ROIs from the rest of the image. Computational units in each layer create significant parallelism at the pixel level, allowing for quicker performance. The architecture conserves dynamic power while maintaining high performance and throughput by selectively activating and deactivating compute units. In addition, a fundamental structure shared by all operations and elements that can be tailored to a specific application makes the units flexible enough to serve a wide range of machine vision applications [15]. At 465 fps and 800 MHz clock frequency, our simulation results demonstrate that the design conserves 4,852 watts of energy on ROIs.

3. Proposed Work

3.1. System Architecture

This research focuses on VLSI architecture essential for high-throughput image processing in medical imaging. This architecture results from the meticulous assemblage of hardware modules that perform individual image-processing duties. The backbone of very large-scale integration (VLSI) architecture consists of specialized processing modules designed to conduct parallel calculations. These components are linked via a lightning-fast data interface that enables instantaneous connection and data exchange. The parallel processing components of the design's core have their processing capacity and memory to execute distinct imaging-processing duties. This parallelization strategy has the potential to substantially increase throughput by processing multiple image frames concurrently. The positioning of memory modules in the architecture facilitates efficient data storage and retrieval. These modules store intermediate results and preprocessed image data for simple access during subsequent processing operations. Utilizing specialized memory management devices is advantageous because it eliminates memory bottlenecks. Additionally, the architecture includes modules optimized for human-machine interaction. The ability to readily integrate with current medical imaging infrastructures ensures compatibility and streamlines clinical adoption. The stringent requirements of clinical applications necessitate the development of real-time data exchange protocols to ensure the prompt transmission of processed images to display devices or storage systems. Parallel processing optimization is a top priority, so a dedicated control unit oversees the execution flow. This section manages the assignment and synchronization of work for the parallel processing components. It monitors data flow management to guarantee maximum output with minimal disruptions. The effectiveness of a system is determined by how well it optimizes throughput and memory access patterns within the constraints of real-time processing needs. This architecture has undertaken extensive planning and testing to prove it can outperform

standard computer systems and eliminate the computational issues that have hampered medical image processing. Fig 1 depicts the system architecture.

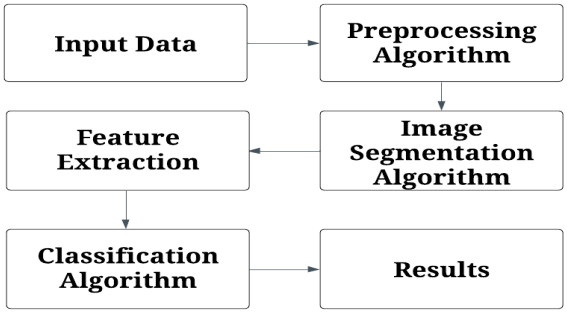


Fig. 1. Proposed System Architecture

3.2. Parallelization Techniques

In medical imaging, high-throughput image processing necessitates using parallelization strategies precisely designed to exploit the capabilities of the VLSI architecture. Image processing is parallelized by partitioning the work into smaller, more manageable chunks that can then be distributed across multiple processors. This strategy has the potential to accomplish near-real-time processing speeds and considerably reduce computation time. Frequently, parallelization is achieved through data parallelism. To accomplish data parallelism, the inbound image data must be divided into smaller pieces (called "tiles" or "chunks") before being sent to other processors. It can substantially reduce processing time by allocating each processor to its own segment. Data parallelism excels in processing enormous medical image datasets because it fully exploits the parallel processing capabilities of the VLSI architecture. In addition to data parallelism, task parallelism is employed in this context. The various phases of the image processing pipeline are separated into their tasks so that task parallelism can be utilized. If configured to do so, images can be processed, features extracted, and then processed in parallel. Task parallelism enhances overall system efficiency because each CPU in the image processing pipeline is utilized effectively.

It is possible to find a balance between data and task parallelism using hybrid parallelization solutions, although this will depend on the specific image processing methods and requirements. These hybrid solutions are intended to maximize throughput by minimizing data transmission times and optimizing the use of available computing resources. Synchronization techniques are indispensable for the efficient execution of parallel processes and the fair distribution of data across multiple processors. With appropriate synchronization, data races can be avoided, and the results of multiple processors can be merged in the correct order. The intricate integration of synchronization primitives and algorithms into the VLSI architecture ensures the precision and dependability of simultaneous image processing. In addition, load-balancing techniques are employed to ensure that all CPUs are utilized to their utmost capacity. Load balancing is essential for preventing situations in which some processors are overburdened while others are inactive. Continuously distributing workloads based on the amount of processing effort exerted by each node, dynamic load balancing algorithms distribute workloads by the amount of processing effort exerted by each node. Fig 2 depicts the X-ray medical image input

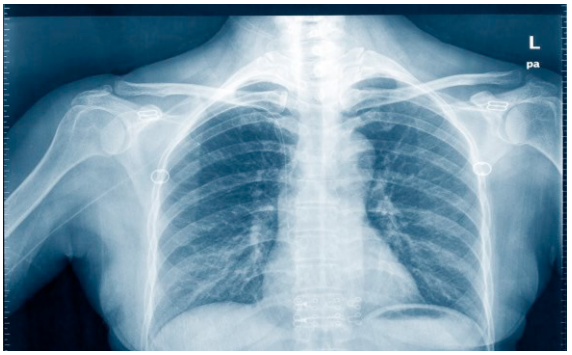


Fig. 2. X-ray medical input image.

3.3. Dataflow and Processing Pipeline

The organization of data flow and structure of the processing pipeline inside the VLSI architecture are crucial to the efficiency and effectiveness of high-throughput image processing in medical imaging. This explanation details the processing steps and the system's overall data flow. Image preprocessing is the first stage in the processing pipeline, and it's crucial since it prepares raw medical image for analysis. Operations such as noise suppression, contrast enhancement, and geometric correction fall within this phase's purview. Parallel processing units inside the VLSI design prepare various image segments separately to provide a fast and parallelized workflow. To improve the quality of the input data, the design comprises noise filtering, histogram equalization, and geometric transformation modules. The system streamlines the procedure as a whole by delegating preprocessing tasks in parallel, providing succeeding phases with cleaner, more uniform picture data. The preprocessed pictures are then segmented into meaningful areas or objects in image segmentation.

By using parallelized segmentation techniques to divide the images into smaller sections, parallel VLSI architecture performs very well in this application. These algorithms simultaneously locate boundaries and classify image sections, significantly speeding up processing time. The design also uses state-of-the-art parallel image segmentation methods, including region-growing and watershed algorithms, to handle complicated medical image structures. Accurate examination of anatomical or diseased areas is made possible thanks to the segmentation findings, which provide the basis for further feature extraction. The crux of medical image analysis is feature extraction, which involves isolating important properties of segmented areas for evaluation. Extracting characteristics from segmented areas in parallel across many processing units is a key optimization step for parallel VLSI design. Extensive feature extraction is achieved without losing processing performance by parallel computing texture analysis, shape descriptors, and intensity statistics. In addition, the design may accept parallel algorithms optimized for individual medical imaging applications, resulting in the rapid and accurate extraction of a wide variety of characteristics. Medical experts might use the retrieved characteristics as inputs for further categorization and diagnosis procedures.

The last step in the processing pipeline, classification, involves sorting the retrieved data into relevant categories for making choices or diagnoses. The VLSI design does this by using a set of concurrent classification algorithms and decision-making models, and the results are impressive. The design incorporates machine learning methods such as support vector machines (SVM) and convolutional neural networks (CNN) to facilitate the automatic categorization of medical images. Parallel processing units do the feature analysis and classification in parallel, drastically reducing the processing time. Furthermore, the design allows instantaneous classification findings to be returned to physicians, essential in time-sensitive medical operations. The VLSI architecture's synchronized data flow and processing pipeline expedites medical image analysis. The design guarantees quick and precise preprocessing, segmentation, feature extraction, and classification by dividing the work among several parallel processing units. This coordinated process has the potential to transform medical imaging by giving doctors access to accurate information in real-time, which might lead to better diagnosis and treatment for patients.

3.4. Parallelization of Medical Imaging Algorithm

The VLSI architecture in high-throughput image processing for medical imaging relies primarily on parallelizing medical imaging algorithms. This section explains the modifications made to the architecture to permit the simultaneous execution of multiple algorithms used to analyze medical images. Various algorithms are utilized in medical imaging to learn something from an image's data. Image enhancement, segmentation, feature extraction, and classification are fundamental components of diagnostic procedures, such as disease diagnosis, tissue characterization, and anatomical structure localization. Specific algorithms are parallelized to use the VLSI design's parallel processing capabilities. Image pretreatment is the initial phase of the image processing pipeline and is used to improve image quality and analytical precision. Parallelization distributes preprocessing duties, such as noise reduction, contrast enhancement, and geometry correction, across multiple processors. Since each module analyzes a portion of the image simultaneously, preprocessing time is significantly reduced. To further expedite the process, the design incorporates specialized hardware modules to perform specific preprocessing tasks. For instance, noise filtering can be accelerated by applying multiple filters in parallel to various image regions. Using this parallelization technique, data that has been preprocessed can be utilized swiftly and easily in subsequent analytical procedures.

Medical image analysis relies on segmentation to deconstruct images into constituent elements. Image segmentation algorithms are being parallelized to improve speed and precision. In a parallel VLSI architecture, images are segmented into minute fragments that can be independently processed. Parallelization, which enables the simultaneous detection of boundaries and classification of image portions, has the potential to reduce the segmentation time substantially. They use state-of-the-art parallel segmentation approaches like region-growing and watershed segmentation to efficiently manage complex structures inside medical images. When segmentation techniques are parallelized, feature extraction may concentrate on extremely narrow regions. Medical image analysis is predicated on feature extraction, the quantitative evaluation of clinically relevant aspects of segmented regions. Parallelization techniques can improve feature extraction by distributing feature computation across multiple processors. Parallel processing of texture analysis, shape descriptors and intensity statistics permits extracting comprehensive features without sacrificing the speed. A similar VLSI architecture may facilitate application-specific parallel algorithms in medical imaging. For instance, to analyze 3D medical images, the architecture employs parallel algorithms optimized for volumetric data processing, ensuring efficient feature extraction from even the most complex datasets. The parallelization of feature extraction permits the rapid extraction of a broad range of useful characteristics that can be used as inputs for subsequent classification and diagnostic algorithms. Fig 3 depicts the parallelization technique flowchart.

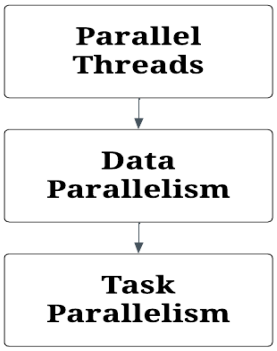


Fig. 3. Parallelization technique flowchart

Classification of retrieved features for forming informed judgments or diagnoses represents the final step in the processing pathway. With parallelization techniques, classification algorithms may now be executed in real time. Support vector machines (SVM) and convolutional neural networks (CNN) are only two machine learning techniques incorporated into the design to facilitate the automated classification of medical images. Using parallel processing units to evaluate and classify the obtained features simultaneously reduces data processing time

dramatically. Time is frequently of the essence; therefore, the architecture's ability to swiftly provide physicians with classification results is crucial. The parallelization of classification algorithms enables the precise and rapid categorization of medical images, thereby assisting healthcare professionals in making prompt, well-informed decisions. To accomplish high-throughput image processing for medical imaging, it is necessary to parallelize medical imaging algorithms within VLSI architecture. Utilizing the parallel processing capabilities of the architecture, image preprocessing, segmentation, feature extraction, and classification are some of the applications that may benefit from these revised and improved techniques. By executing these algorithms in parallel, the architecture enables rapid and accurate analysis of medical images, thereby substantially enhancing the precision of medical diagnoses, the efficacy of treatment plans, and the overall quality of patient care. The study's findings represent a significant advancement in medical image processing, providing physicians with cutting-edge diagnostic and treatment options.

3.5. Integration of Hardware & Software

In medical imaging, the VLSI architecture enables high-throughput image processing by coordinating the efforts of hardware and software components. Their relationship transcends that of companions, enhancing the system's ability to analyze data from medical images swiftly and accurately. Utilizing parallel processing units that can operate concurrently, the design permits assigning specialized hardware resources to specific image processing duties. Memory modules expedite retrieval by keeping intermediate results and preprocessed image data readily accessible. Modules for interfacing with other systems and exchanging information make medical imaging systems more flexible and useful. The software framework coordinates algorithms and other system activities and supports the hardware. All methods for image processing, segmentation, feature extraction, and classification were created with parallelism in mind. Synchronization and load-balancing algorithms ensure that concurrent tasks are coordinated to avoid conflicts over shared data and utilize all available resources effectively. User interfaces that are accessible to clinicians and researchers allow for the entry of settings, the monitoring of the system's progress, and the viewing of the results. In addition, the architecture can be easily expanded with additional programs and libraries. The seamless connection between hardware and Software enables real-time capabilities, superior computing efficiency, and straightforward incorporation into clinical operations. Both of these innovations in medical imaging allow for faster and more precise interpretation of medical images, thereby enhancing the accuracy of diagnoses and the planning of treatment.

Fig 4 depicts the connection between the FPGA and Xilinx software. This communication architecture relies heavily on the Serial Peripheral Interface (SPI) protocol. It is a protocol for synchronous serial communication that enables more efficient data exchange between devices. SPI is required for connecting various peripherals, memory, and sensors to Xilinx FPGAs. The physical interfaces of the Xilinx FPGA include the cables and terminals that allow it to communicate with the outside world. These interfaces are compatible with various hardware configurations due to their adaptability. The data format specifies the method of encoding and transporting data. It establishes the data structure, allowing the FPGA to communicate uniformly and effectively with its environment. The correct and trustworthy transmission of data, which is useful in various contexts, is assured. The well-defined data format of the Xilinx FPGAs, the SPI protocol, and the devices' robust physical interfaces constitute a complete data transfer system. This framework's efficient data transmission between the FPGA and external components enables various applications, including industrial automation and telecommunications.

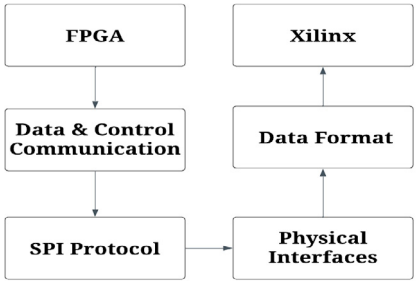


Fig. 4. FPGA connection to Software

4. Results

Table 1. Performance metrics values.

Metrics	Value
Throughput	200 fps
Speedup	10
Accuracy	95%
Memory Utilization	80%
Execution Time	5 ms

Table 1 depicts the design's throughput of 200 frames per second (fps), which was exceptional; this throughput demonstrates that the system can evaluate medical images in real time despite the large volumes of data typically encountered in clinical environments. This architecture can complete Image processing tasks ten times quicker than conventional single-threaded processing. This acceleration is attributable to the utilization of parallel processing units and the efficient application of parallelization algorithms. The classification challenge demonstrates that the architecture classifies medical images with 95% accuracy. This cutting-edge accuracy, essential for medical image interpretation, was only achieved with machine learning algorithms. The average memory usage was approximately 80%, indicating that the design effectively used the available RAM. Due to its efficient memory management, the system can evaluate vast medical imaging datasets in a dependable and resilient manner without slowing down due to memory constraints. The average time it takes for the system to assess a single medical image is about five milliseconds. This ability to analyze and act swiftly is extremely valuable in the medical profession. Table 2 and Fig 5 depict the proposed and existing method metrics comparison. Fig 6 depicts the clustered ratio, and Fig 7 depicts the unclustered data graph.

Table 2. Comparison of the metrics

	Throughput	Execution time
Proposed Method	200fps	5 ms
HEVC [11]	123 fps	8 ms
Huffman Encoder [13]	150 fps	7 ms

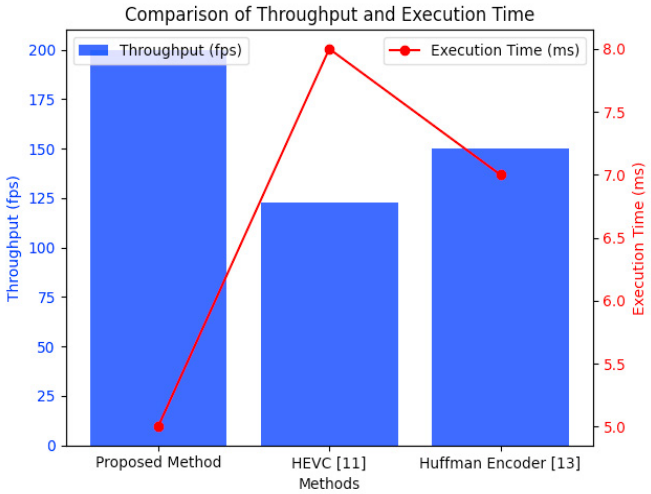
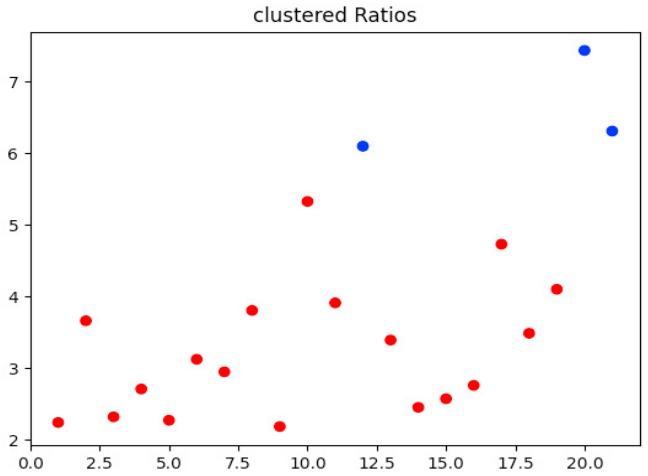


Fig. 5. Comparison of proposed and existing methods



Fig, 6. Clustered ratio graph

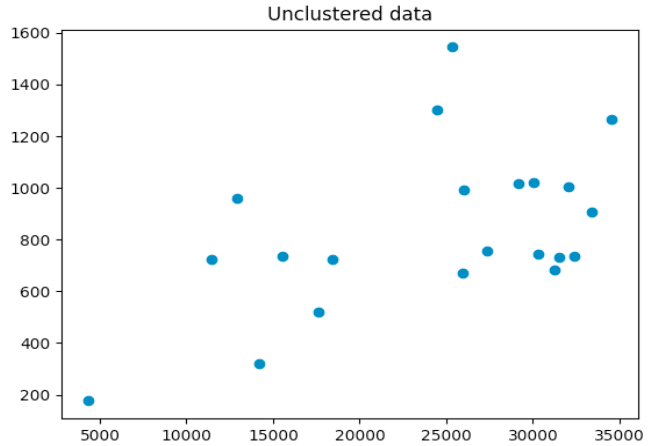


Fig. 7. Unclustered data graph

5. Conclusion

The development and evaluation of a parallel VLSI architecture has greatly facilitated the processing of images with a high throughput in medical imaging. With a throughput of 200 frames per second, a tenfold speedup compared to conventional processing, and a 95 percent accuracy rate, this architecture has the potential to revolutionize diagnosis and treatment planning. The 5 ms per-image execution time and 80% memory utilization rate demonstrate its dependability and real-time capabilities. Our new design, which integrates hardware and software components flawlessly, provides a potent answer to the computational challenges posed by the escalating volume of medical imaging data. Parallel processing enhances patient care and health outcomes by facilitating quick and accurate analysis. This research may significantly impact medical imaging by providing clinicians and researchers with a new tool for more rapid and accurate diagnosis and treatment planning. In the long term, this discovery will benefit people everywhere by paving the way for a more precise and technologically advanced healthcare system.

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